SOLUTIONS: DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

1. a) Suppose that Δ and S are stable. Then the feedback loop is stable if $\|\Delta S\|_\infty < 1$.

b) It is clear that we require $K(s)$ to be internally stabilising.

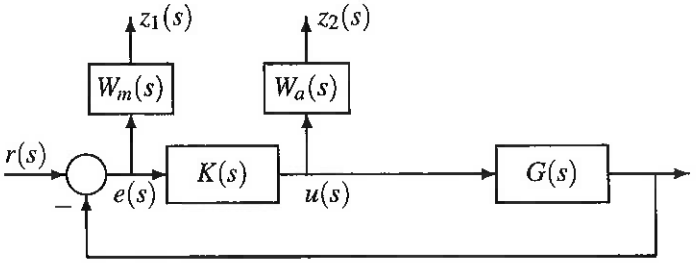
i) Suppose that $\Delta_a = 0$ and let the input to Δ_m be ε_2 while the output from Δ_m be δ_2. Then a calculation shows that $\varepsilon_2 = (I+GK)^{-1}\delta_2$. Using the small gain theorem, to satisfy the first design requirement, it is sufficient that $\left\|\Delta_m(j\omega)\,(I+G(j\omega)K(j\omega))^{-1}\right\| < 1, \forall\omega$. This can be satisfied if $\left\|W_m(I+GK)^{-1}\right\|_\infty < 1$, where $W_m = w_m I$.

ii) An analogous procedure shows that to satisfy the second design requirement, it is sufficient that $\left\|\Delta_a(j\omega)K(j\omega)\,(I+G(j\omega)K(j\omega))^{-1}\right\| < 1, \forall\omega$. This can be satisfied if $\left\|W_a K\,(I+GK)^{-1}\right\|_\infty < 1$, where $W_a = w_a I$.

Thus, to satisfy both design requirements, it is sufficient that

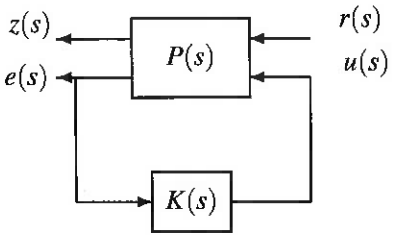
$$\left\| \begin{bmatrix} W_m(I+GK)^{-1} \\ W_a K\,(I+GK)^{-1} \end{bmatrix} \right\|_\infty < 1.$$

c) All the requested signals are shown in the block diagram shown below.



d) The corresponding generalised regulator formulation is to find an internally stabilising $K(s)$ such that $\|\mathcal{F}_l(P,K)\|_\infty < 1$ where

$$z(s) = \begin{bmatrix} z_1(s) \\ z_2(s) \end{bmatrix},\ P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} = \left[\begin{array}{c|c} W_m(s) & -W_m(s)G(s) \\ 0 & W_a(s) \\ \hline I & -G(s) \end{array}\right].$$



e) Since $e(s) = (I+G(s)K(s))^{-1}r(s)$, it follows that the tracking properties are determined by the weighting function $w_m(j\omega)$: the smaller $|w_m^{-1}(j\omega)|$ the better the tracking at ω.

 Copyright Imperial College London

2. a) i) The $(1,1)$ block of the inequality gives $A^TP+PA \prec 0$. Let $Az=\lambda z$, $z\neq 0$. Then multiplying the inequality from the left by z^T and from the right by z gives $(\lambda+\bar{\lambda})z^TPz<0$. Since $P\succ 0$ it follows that $z^TPz>0$ and it follows that $\lambda+\bar{\lambda}<0$ so that A is stable.

ii) Since A is stable, $\|H\|_\infty<\gamma$ if and only if, with $x(0)=0$, $J:=\int_0^\infty[y^Ty-\gamma^2u^Tu]dt<0$, for all $u(t)$ such that $\|u\|_2<\infty$. If $\|u\|_2$ is bounded, then $\lim_{t\to\infty}x(t)=0$. Now, $\int_0^\infty\frac{d}{dt}[x^TPx]dt=x(\infty)^TPx(\infty)-x(0)^TPx(0)=0$. So,

$$0=\int_0^\infty\left(\dot{x}^TPx+x^TP\dot{x}\right)dt=\int_0^\infty[x^T(A^TP+PA)x+x^TPBu+u^TB^TPx]dt.$$

Use $y=Cx+Du$ and add the last expression to J

$$\begin{aligned}J &= \int_0^\infty[x^T(A^TP+PA+C^TC)x+2x^T(PB+C^TD)u+u^T(D^TD-\gamma^2I)u]dt\\ &= \int_0^\infty\begin{bmatrix}x^T & u^T\end{bmatrix}\overbrace{\begin{bmatrix}A^TP+PA+C^TC & PB+C^TD\\ B^TP+D^TC & D^TD-\gamma^2I\end{bmatrix}}^{M}\begin{bmatrix}x\\ u\end{bmatrix}dt.\end{aligned}$$

It follows that $J<0$, and so $\|H\|_\infty<\gamma$, if $M\prec 0$. This proves the result.

iii) Now,

$$M=\begin{bmatrix}A'P+PA & PB\\ B'P & -\gamma^2I\end{bmatrix}-\begin{bmatrix}C'\\ D'\end{bmatrix}(-I)\begin{bmatrix}C & D\end{bmatrix}$$

It follow that

$$H(s)\text{ is stable and }\|H\|_\infty<\gamma\Leftrightarrow\begin{bmatrix}A'P+PA & PB & C'\\ B'P & -\gamma^2I & D'\\ C & D & -I\end{bmatrix}\prec 0.$$

b) i) Substituting $u=Le$ into the state equation gives

$$\dot{x}=\underbrace{(A+LC)}_{A_c}x+\underbrace{\begin{bmatrix}B & L\end{bmatrix}}_{B_c}w,\qquad e=\underbrace{C}_{C_c}x+\underbrace{\begin{bmatrix}0 & I\end{bmatrix}}_{D_c}w.$$

It follows that $T_{ew}(s)=D_c+C_c(sI-A_c)^{-1}B_c$.

ii) Using the results of part (aii), by replacing A, B, C and D by A_c, B_c, C_c and D_c, we have that there exists a feasible L if there exists $P=P'\succ 0$ such that

$$\begin{bmatrix}(A+LC)'P+P(A+LC)+C'C & PB & PL+C'\\ B'P & -\gamma^2I & 0\\ L'P+C & 0 & (1-\gamma^2)I\end{bmatrix}\prec 0.$$

Noting that the only nonlinearity is due to the product PL, we define $Z=PL$ and so there exists a feasible L if there exists $P=P'\succ 0$ and Z such that

$$\begin{bmatrix}A'P+PA+ZC+C'Z'+C'C & PB & Z+C'\\ B'P & -\gamma^2I & 0\\ Z'+C & 0 & (1-\gamma^2)I\end{bmatrix}\prec 0.$$

 Copyright Imperial College London

3. a) Let $V = x'Px$ and set $u = -Fx$. Provided that $P = P' \succ 0$ and $\dot{V} < 0$ along closed-loop trajectories, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Px + x'P\dot{x} = x'\left(A'P + PA - F'B'P - PBF\right)x.$$

Integrating from 0 to ∞ and using $x(\infty) = 0$,

$$\int_0^\infty x'\left(A'P + PA - F'B'P - PBF\right)xdt = -x_0'Px_0.$$

Using the definition of J, adding the last equation and completing a square:

$$J = x_0'Px_0 + \int_0^\infty \{x'[A'P + PA + I - PBB'P]x + \|(F - B'P)x\|^2\}dt.$$

Since the last term is always nonnegative, it follows that the minimizing value of F is given by $F = B'P$. We can set the term in square brackets to zero provided P satisfies the Riccati equation,

$$A'P + PA + I - PBB'P = 0.$$

It follows that the minimum value of J is $x_0'Px_0$.

b) We need to prove that $A_c := A - BB'P$ is stable. The Riccati equation can be written as $A_c'P + PA_c + I + PBB'P = 0$. Let $\lambda \in \mathscr{C}$ be an eigenvalue of A_c and $z \neq 0$ be the corresponding eigenvector. Pre– and post–multiplying the Riccati equation be z' and z respectively gives $(\lambda + \bar{\lambda})z'Pz + z'z + z'PBB'Pz = 0$. Since $P \succ 0$ and $z \neq 0$, $z'Pz > 0$, $z'z > 0$ and $z'PBB'Pz \geq 0$. It follows that $\lambda + \bar{\lambda} < 0$ and the closed loop is stable.

c) The loop-gain is given by $L(s) = F(sI - A)^{-1}B$.. Let λ be a zero of $L(s)$ so that,

$$\begin{bmatrix} A - \lambda I & B \\ F & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

for some v_1, v_2 with $v_1 \neq 0$ since B has full column rank. Then,

$$Av_1 = \lambda v_1 - Bv_2,$$

and,

$$Fv_1 = 0 \Rightarrow B'Pv_1 = 0.$$

Consider the Riccati equation. Then,

$$\begin{aligned} 0 &= v_1'[A'P + PA - PBB'P + I]v_1 \\ &= (\bar{\lambda}v_1' - v_2'B')Pv_1 + v_1'P(\lambda v_1 - Bv_2) + v_1'v_1 \\ &= (\bar{\lambda} + \lambda)\underbrace{v_1'Pv_1}_{>0} + \underbrace{v_1'v_1}_{>0} \end{aligned}$$

Hence $\lambda + \bar{\lambda} < 0$ and $L(s)$ is minimum-phase.

 Copyright Imperial College London

4. a) The generalized regulator formulation is given by

$$\begin{bmatrix} z(s) \\ y(s) \end{bmatrix} = P(s) \begin{bmatrix} w(s) \\ u(s) \end{bmatrix}, \; u(s) = Fy(s), P(s) = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \stackrel{s}{=} \left[\begin{array}{c|c|c} A & B & B \\ \hline I & 0 & 0 \\ 0 & 0 & I \\ \hline I & 0 & 0 \end{array}\right].$$

b) Let $J = \|z\|_2^2 - \gamma^2 \|w\|_2^2$ denote the cost function. Let $V = x'Xx$ and set $u = Fx$. Provided that $X = X' \succ 0$ and $\dot{V} < 0$ along the closed-loop trajectory, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Xx + x'X\dot{x} = x'\left(A'X + XA + F'B'X + XBF\right)x + x'XBw_1 + w_1'B'Xx.$$

Integrating from 0 to ∞ and using $x(\infty) = 0$,

$$-x_0'Xx_0 = \int_0^\infty \left[x'\left(A'X + XA + F'B'X + XBF\right)x + x'XBw_1 + w_1'B'Xx\right]dt.$$

Using the definition of J, adding the last equation, and completing two squares,

$$J = x_0'Xx_0 + \int_0^\infty \left\{x'\left[A'X + XA + I - X\left(BB' - \gamma^{-2}BB'\right)X\right]x + \|Zx\|^2 - \|v\|^2\right\}dt$$

where $Z = F + B'X$ and $v = \gamma w - \gamma^{-1}B'Xx$.

c) The requirement $\|H\|_\infty \le \gamma$ is equivalent to $J \le 0$ when $x_0 = 0$. Thus sufficient conditions for $J \le 0$ are $Z = 0$ and the existence of X such that

$$A'X + XA + I - X\left(BB' - \gamma^{-2}BB'\right)X = 0, \qquad X = X' \succ 0.$$

Setting $Z = 0$, the control policy is $F = -B'X$. Setting $v = 0$, the worst case disturbance is $w^* = \gamma^{-2}B'X$. The closed-loop with $u = Fx$ and $w = w^*$ is $\dot{x} = A_c x$ where $A_c = A - \left(BB' - \gamma^{-2}BB'\right)$ and a further condition is $Re\ \lambda_i(A_c) < 0, \; \forall\, i$.

d) Using the expression for J in Part b) and the solution in Part c) with $\gamma \to \infty$, it follows that $J \le x_0^T X x_0$ and the Riccati equation becomes

$$A'X + XA + I - XBB'X = 0, \qquad X = X' \succ 0.$$

Thus, as $\gamma \to \infty$, the solution of the $\mathscr{H}_\infty$ regulator converges to that of the LQR in Question 3.

 Copyright Imperial College London